

Program–3 Code a Dockerized Python Flask Application

Folder structure:

```
program-3/  
| — Dockerfile  
| — app.py  
| — requirements.txt
```

STEP 1: Write Flask Application (app.py)

nano app.py

```
GNU nano 7.2 app.py  
from flask import Flask  
  
app = Flask(__name__)  
  
@app.route("/")  
def home():  
    return "Hello from Simple Flask Docker!"  
  
if __name__ == "__main__":  
    app.run(host="0.0.0.0", port=5000)
```

STEP 2: Create requirements.txt

nano requirements.txt

```
GNU nano 7.2 requirements.txt  
Flask==2.3.3
```

STEP 3: Create Dockerfile

nano Dockerfile

```
GNU nano 7.2 Dockerfile
FROM python:latest

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 5000

CMD ["python", "app.py"]
```

STEP 4: Build Docker Image

docker build -t program-3 .

STEP 5: Run Docker Container

docker run -p 5000:5000 program-3

```
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program3$ docker build -t program-3 .
[+] Building 291.0s (11/11) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 197B                                0.0s
=> [internal] load metadata for docker.io/library/python:latest    9.0s
=> [auth] library/python:pull token for registry-1.docker.io       0.0s
=> [internal] load .dockerignore                                    0.0s
=> => transferring context: 2B                                       0.0s
=> [internal] load build context                                    0.0s
=> => transferring context: 480B                                      0.0s
=> [1/5] FROM docker.io/library/python:latest@sha256:6d58c1a9444bc2664f0fa20c43a592fcdcb2698eb9a9c32257516538a2746c19a 275.7s
=> resolve docker.io/library/python:latest@sha256:6d58c1a9444bc2664f0fa20c43a592fcdcb2698eb9a9c32257516538a2746c19a 0.0s
=> sha256:6d58c1a9444bc2664f0fa20c43a592fcdcb2698eb9a9c32257516538a2746c19a 10.95kB / 10.95kB 0.0s
=> sha256:281b80c799ded5b9a390d2e8c59930db01ee395ab809dd34259897c660751f4e 49.29MB / 49.29MB 151.9s
=> sha256:15f14138abe4f09d3ef3953105144728056046ae469ae21e8e42749bacd67cca 25.61MB / 25.61MB 17.1s
=> sha256:378c64c4458002be86f2d86c5768ae9ec8cfff69afac7d1444e50206dc4566fa9 67.78MB / 67.78MB 154.5s
=> sha256:37c709cf7651a764f8c33f574fe4771be54f77cf626c3b2572e26493ca465e3b1 2.32kB / 2.32kB 0.0s
=> sha256:1edf53fe0281306069b0c05371a2ad9c5ac12e7474dc0cd6c4902c7eb33a9c9 6.49kB / 6.49kB 0.0s
=> sha256:02e37abc533a09f86edcf3602251a61a6208ddaa131e1a1d5757e5f94a54d645 235.98MB / 235.98MB 268.2s
=> extracting sha256:281b80c799ded5b9a390d2e8c59930db01ee395ab809dd34259897c660751f4e 1.4s
=> sha256:b1741a62cceed4e959c504d7a6a25e533c5783994c386b24d7b31a646163283d 6.08MB / 6.08MB 218.9s
=> extracting sha256:15f14138abe4f09d3ef3953105144728056046ae469ae21e8e42749bacd67cca 0.6s
=> sha256:ad107facb26b82200d59d550cbbab6af9cf38c971f00a43e3d3cda71ffe76a 29.40MB / 29.40MB 261.4s
=> extracting sha256:378c64c4458002be86f2d86c5768ae9ec8cfff69afac7d1444e50206dc4566fa9 2.7s
=> sha256:1227bf08bd420f8324f592cf370b34de4c11dd57aba693e8f3ec9ccbd8d468d 249B / 249B 220.8s
=> extracting sha256:02e37abc533a09f86edcf3602251a61a6208ddaa131e1a1d5757e5f94a54d645 6.3s
=> extracting sha256:b1741a62cceed4e959c504d7a6a25e533c5783994c386b24d7b31a646163283d 0.2s
=> extracting sha256:ad107facb26b82200d59d550cbbab6af9cf38c971f00a43e3d3cda71ffe76a 0.6s
=> extracting sha256:1227bf08bd420f8324f592cf370b34de4c11dd57aba693e8f3ec9ccbd8d468d 0.0s
=> [2/5] WORKDIR /app                                              0.7s
=> [3/5] COPY requirements.txt .                                    0.0s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt       5.2s
=> [5/5] COPY . .                                                  0.0s
=> exporting to image                                              0.2s
=> => exporting layers                                             0.2s
=> => writing image sha256:e74e02e8d91045cb80e42449afbc60e9912885acd89de2ac3bd709b8ecc1f2e 0.0s
=> => naming to docker.io/library/program-3                        0.0s
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program3$ docker run -p 5000:5000 program-3
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Press CTRL+C to quit
```

STEP 6: Test in Browser

Open browser:

`http://localhost:5000`

